Predictive Maintenance of Power Transformers Using Dissolved Gas Analysis and Machine Learning

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Abstract

Power transformers are critical components of electrical power systems, and their reliable operation is essential for maintaining a continuous and stable supply of electricity. Electrical and thermal faults, including overheating, insulation deterioration, partial discharge, and arcing, can develop during transformer operation and may result in equipment damage and unexpected power interruptions if they are not detected early. Dissolved Gas Analysis (DGA) is widely used for condition monitoring of oil-immersed transformers because the concentrations of gases dissolved in transformer oil provide information about internal conditions and possible fault types. This study investigates machine-learning techniques for DGAbased transformer fault classification. Five dissolved-gas measurements and their corresponding condition labels are used as model inputs and targets. Six classifiers—Random Forest (RF), XGBoost (XGB), K-Nearest Neighbors (KNN), Artificial Neural Network (ANN), Support Vector Machine (SVM), and Long Short-Term Memory (LSTM)—are compared. Data cleaning, feature preparation, label encoding, scaling where required, and an 80:20 train–test split are considered. Model performance is assessed using accuracy, precision, recall, specificity, F1score, Matthews Correlation Coefficient (MCC), and confusion matrices. In the reported experiment, XGBoost achieved the highest accuracy of 86.67%, followed by Random Forest at 85.16% and KNN at 83.23%. The results indicate the potential of combining DGA measurements with machine learning to support transformer condition monitoring and early fault identification. Further validation on larger and independent field datasets is required before practical deployment.

Index Terms— Dissolved Gas Analysis, transformer fault diagnosis, predictive maintenance, machine learning, fault classification, condition monitoring.

1 Introduction

1.1 Background and Context

Power transformers are essential components of the electricity supply chain, transferring electrical energy between voltage levels. Their reliable operation is important for maintaining continuity and stability of supply. Transformer faults and prolonged outages can cause financial losses, service interruptions, and safety concerns. Continuous condition monitoring is therefore important for identifying developing abnormalities and supporting timely maintenance decisions. The insulation system of an oil-immersed transformer primarily consists of insulating paper and oil. These materials are exposed to electrical, thermal, and mechanical stresses throughout the transformer's operating life. Deterioration of the insulation system can reduce transformer performance and eventually lead to internal faults. Thermal

stress, electrical discharge, and insulation degradation can change the condition of insulating materials and generate characteristic gaseous products within the transformer oil. Dissolved Gas Analysis is a non-destructive diagnostic technique in which the type and concentration of gases dissolved in insulating oil are analysed to estimate the internal condition of a transformer [1, 2, 3]. Hydrogen, methane, ethane, ethylene, and acetylene can be associated with different thermal and electrical fault conditions. The quantity and composition of these gases provide useful features for transformer fault diagnosis.

1.2 Problem Statement

Although DGA provides valuable information for transformer condition assessment, reliable fault classification can be challenging because different fault conditions may produce overlapping gas patterns. Variations in gas concentrations, operating conditions, and fault severity can complicate direct interpretation of measurements. Conventional diagnostic approaches may not always clearly distinguish multiple fault categories. This study uses dissolved-gas measurements as input features for multiclass transformer fault classification. The objective is to compare multiple machine-learning approaches under a consistent experimental protocol and to examine their strengths and limitations using several performance metrics.

1.3 Research Objectives and Contributions

The main objectives are: • to represent transformer condition using characteristic dissolved-gas measurements; • to assess data quality and prepare a consistent feature and label representation; • to compare RF, XGBoost, KNN, ANN, SVM, and LSTM for multiclass classification; • to evaluate the classifiers using accuracy, precision, recall, specificity, F1-score, MCC, and confusion matrices; and • to outline how model predictions could support condition assessment and maintenance prioritisation.

1.4 Paper Organization

Section II summarises the DGA-based diagnostic context. Section III describes the gas features, transformer condition

Table 1: DGA Features and Their Diagnostic Relevance

Gas	**Diagnostic relevance**
Hydrogen (H_2)	Provides information related to electrical discharge activity and discharge-related classes.
Methane (CH_4)	Contributes to identifying thermal and electrical fault patterns through its concentration relative to other gases.
Ethane (C_2H_6)	Provides information associated with thermal decomposition and helps distinguish fault conditions.
Ethylene (C_2H_4)	Acts as a discriminative feature for thermal-fault patterns, particularly when considered with other hydrocarbon gases.
Acetylene (C_2H_2)	Provides an important feature for identifying high-energy electrical discharge and arcing-related conditions.

classes, and dataset representation. Section IV presents the preprocessing and machine-learning methodology. Section V discusses the reported results. Section VI describes a predictivemaintenance framework, limitations, and future work. Section VII concludes the paper.

2 Dissolved Gas Analysis And Transformer

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2.1 Dissolved Gas Analysis

When an oil-immersed transformer experiences abnormal electrical or thermal stress, the insulating oil and solid insulation can decompose and produce characteristic gases. The measured concentrations of these gases provide information about the condition of the transformer. DGA can therefore be formulated as a data-driven classification problem, with observed gas concentrations as input variables and the corresponding transformer condition as the target class. Five gases are used in this study: hydrogen (H2), methane (CH4), ethane (C2H6), ethylene (C2H4), and acetylene (C2H2). Each gas provides complementary information about possible reactions and fault mechanisms. Hydrogen is often associated with electrical discharge activity; hydrocarbon gases can indicate thermal decomposition; and acetylene is relevant to high-energy discharge and arcing conditions. No single gas concentration uniquely identifies every fault type, so the joint distribution and relative variation of gases are important.

2.2 Feature and Target Representation

For observation i, the input feature vector is xi = [H2,i, CH4,i, C2H6,i, C2H4,i, C2H2,i]T . (1) The associated transformer condition is represented by target label yi. The complete dataset is D = {(xi, yi)}N i=1, (2) where N is the total number of observations. The task is represented as a supervised multiclass classification function: f : R5 → {1, 2, . . . , 7}. (3) The seven condition classes considered are normal condition, partial discharge, low-energy discharge, high-energy discharge, low-temperature overheating, middle-temperature overheating, and high-temperature overheating. Classification depends on the combined gas pattern rather than the concentration of any one gas.

3 Machine-Learning Methodology

3.1 Proposed Diagnostic Framework

The proposed framework formulates transformer fault diagnosis as supervised learning using DGA measurements. Electrical and thermal stresses can generate different combinations of dissolved gases. A classifier is trained to learn patterns in these measurements and estimate the corresponding transformer condition. The overall workflow is: DGA acquisition → data-quality assessment → preprocessing → feature representation → train/test partitioning → model training and selection → fault classification.

3.2 Dataset Description and Data Quality

Before model training, the dataset should be examined for missing values, duplicate observations, invalid measurements, inconsistent class labels, and irrelevant variables. Measurements that are physically inconsistent with the feature definition should be reviewed. Extreme gas concentrations should not automatically be removed, because they may represent genuine severe electrical or thermal faults and contain useful diagnostic information. Target labels are checked to ensure that each observation is consistently assigned to one of the seven condition classes. Categorical labels are encoded into numerical representations where required by the selected algorithm. Exploratory data analysis can use descriptive statistics, histograms, box plots, and correlation analysis to inspect feature ranges, skewness, relationships among gases, and the representation of the target classes.

3.3 Feature Scaling and Transformation

Feature scaling is important for algorithms that rely on distances or feature magnitudes. KNN uses distance-based similarity, while SVM and ANN can also be sensitive to differences in feature scale. Standardisation may be applied using training-set statistics: zij = xij - µj σj , (4) where µj and σj are the mean and standard deviation of feature j calculated from the training data. The same transformation is then applied to validation and test data. Tree-based methods such as RF and XGBoost generally do not require standardisation, but all models use the same underlying five-gas feature definition for a fair comparison. Potential outliers are evaluated in the context of data quality and engineering relevance. A very high concentration may indicate a serious fault rather than a measurement error; therefore, removal or transformation should be justified rather

than applied automatically. Logarithmic transformations or DGA-informed ratios may be investigated for highly skewed features, provided that the transformation is fitted without using held-out test information.

3.4 Train–Test Strategy and Data Leakage Prevention

The dataset is divided into training and testing subsets using an 80:20 split. Stratified partitioning is preferred to preserve the relative representation of the seven target classes. The training subset is used for model fitting, feature-processing decisions, and hyperparameter selection. The test subset remains isolated until final evaluation. Cross-validation is performed within the training data where applicable. Scaling parameters, imputation values, and featureselection decisions must be estimated using training data only. Applying preprocessing statistics calculated from the entire dataset before splitting can cause data leakage and produce overly optimistic performance estimates. Where multiple measurements originate from the same transformer or experimental source, grouped partitioning should be considered to reduce the chance of related observations appearing in both training and test sets.

3.5 Machine-Learning Classifiers

Six machine-learning approaches are considered to provide a comparison across different learning paradigms. 1) Random Forest: Random Forest is an ensemble of decision trees trained using randomised subsets of observations and features [4]. Individual tree predictions are aggregated to obtain the final multiclass prediction. RF can model nonlinear relationships and interactions among dissolved gases without requiring an explicit analytical relationship between the gas concentrations and transformer condition. 2) XGBoost: XGBoost is a gradient-boosting method in which decision trees are constructed sequentially, with each successive tree reducing errors produced by the preceding ensemble [5]. Regularisation helps control model complexity. Important hyperparameters include tree depth, learning rate, number of estimators, and regularisation parameters. The method is suitable for structured data in which gas concentrations may interact nonlinearly. 3) K-Nearest Neighbors: KNN classifies a test observation according to the labels of nearby training observations in feature space [7]. It provides a similarity-based interpretation: a new transformer gas signature is classified according to its proximity to previously observed signatures. Because distance calculations are sensitive to feature magnitude, appropriate scaling is applied before KNN training. 4) Support Vector Machine: SVM constructs decision boundaries by maximising the separation margin between classes [6]. Kernel functions can represent nonlinear boundaries when classes are not linearly separable in the original feature space. Kernel selection, regularisation, and feature scaling are important considerations during model optimisation. 5) Artificial Neural Network: An ANN receives the DGA features through an input layer, learns nonlinear representations through one or more hidden layers, and produces an output corresponding to the transformer-condition classes. Network parameters are optimised using a classification loss function.

Regularisation and early stopping can help reduce overfitting, particularly when the available dataset is limited. 6) Long Short-Term Memory: LSTM is a recurrent neuralnetwork architecture designed to learn dependencies in sequential data [8]. Its relevance to transformer diagnosis is strongest when repeated DGA measurements are available over time and the evolution of gas concentrations contains diagnostic information. If observations are independent and have no genuine temporal ordering, the advantage of LSTM is limited; it should therefore be interpreted as a comparative model unless a meaningful sequence representation is available.

3.6 Hyperparameter Optimisation

Models can be implemented using established machine-learning libraries such as Scikit-learn [9]. Model hyperparameters are selected using the training data, with cross-validation where applicable. Representative parameters include the number and depth of trees for RF, learning rate and tree depth for XGBoost, neighbourhood size for KNN, kernel and regularisation parameters for SVM, and architecture and learning rate for ANN. Sequence-related parameters may be considered for LSTM when appropriate. Grid search or random search can be used depending on the size of the search space. Model selection should rely on training-only validation results; the independent test set is reserved for final evaluation.

4 Evaluation Metrics

The six classifiers are evaluated using accuracy, precision, recall, specificity, F1-score, MCC, and confusion matrices. For a one-versus-rest evaluation of an individual class, TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively. The Matthews Correlation Coefficient is included because it provides a balanced summary of classification quality; its interpretation alongside other metrics is discussed in the literature [10].

4.1 Accuracy

Accuracy measures the proportion of all observations that are correctly classified.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{5}$$

Accuracy provides an overall measure but can be misleading when class frequencies are imbalanced.

4.2 Precision

Precision measures the proportion of positive predictions for a class that are correct.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{6}$$

Higher precision indicates fewer false-positive assignments to the considered class.

4.3 Recall

Recall, also known as sensitivity or true-positive rate, measures the proportion of actual positive observations correctly identified.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{7}$$

Recall is important in transformer fault diagnosis because a low value can indicate that actual fault observations are assigned to another condition.

4.4 Specificity

Specificity measures the proportion of negative observations correctly rejected.

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{8}$$

High specificity indicates that observations belonging to other conditions are less likely to be incorrectly assigned to the considered class.

4.5 F1-Score

The F1-score is the harmonic mean of precision and recall.

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{9}$$

For multiclass classification, class-wise scores can be calculated using a one-versus-rest formulation and macro-averaged so that each class contributes equally.

4.6 Matthews Correlation Coefficient

For binary classification, the Matthews Correlation Coefficient (MCC) is defined as

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{10}$$

For multiclass classification, MCC is computed from the complete confusion matrix. It accounts for correct and incorrect predictions across classes and is useful when class distributions are not perfectly balanced.

4.7 Confusion Matrix

A confusion matrix provides a class-by-class view of classifier behaviour. Rows represent actual classes and columns represent predicted classes. Diagonal elements correspond to correctly classified observations, while off-diagonal elements represent misclassifications. This is particularly important for DGA because distinct electrical and thermal faults can produce overlapping gas signatures.

5 Results And Discussion

5.1 Accuracy Comparison

Fig. 1 compares the reported accuracy of the six classifiers. XGBoost achieved the highest reported accuracy, followed by Random Forest and KNN.

5.2 Confusion Matrices

The confusion matrices show correct classifications along the diagonal and misclassifications in the off-diagonal cells. They provide class-level detail that complements the overall accuracy results.

5.3 Overall Model Performance

Fig. 8 compares the reported accuracy, precision, recall, specificity, and F1-score of the six classifiers.

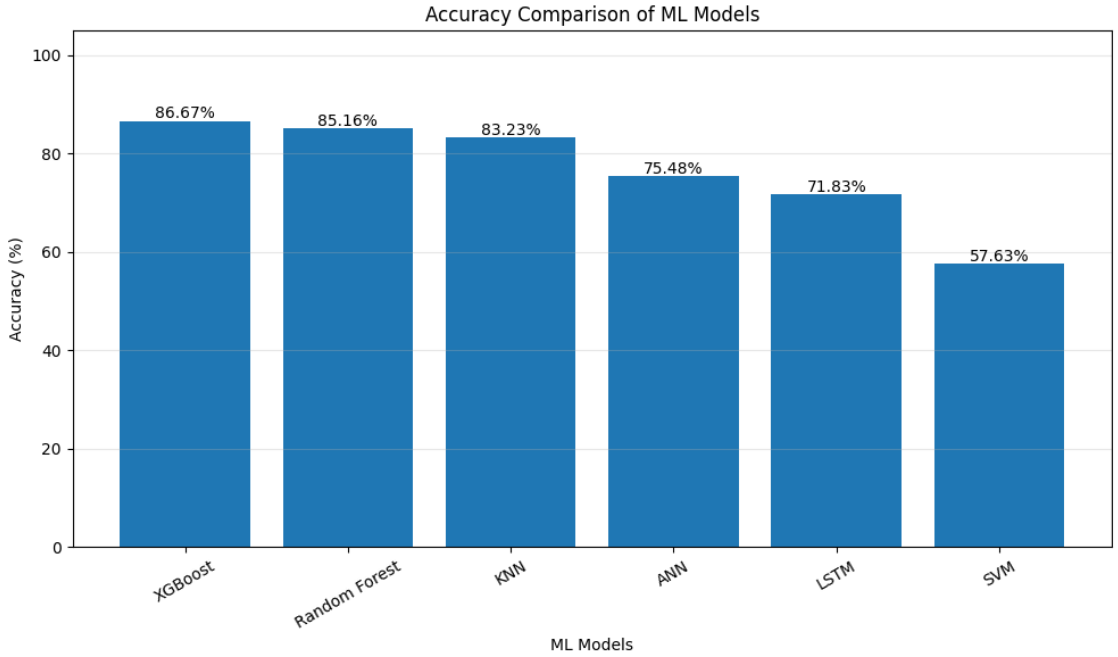


Figure 1: Accuracy comparison of the six machine-learning models.

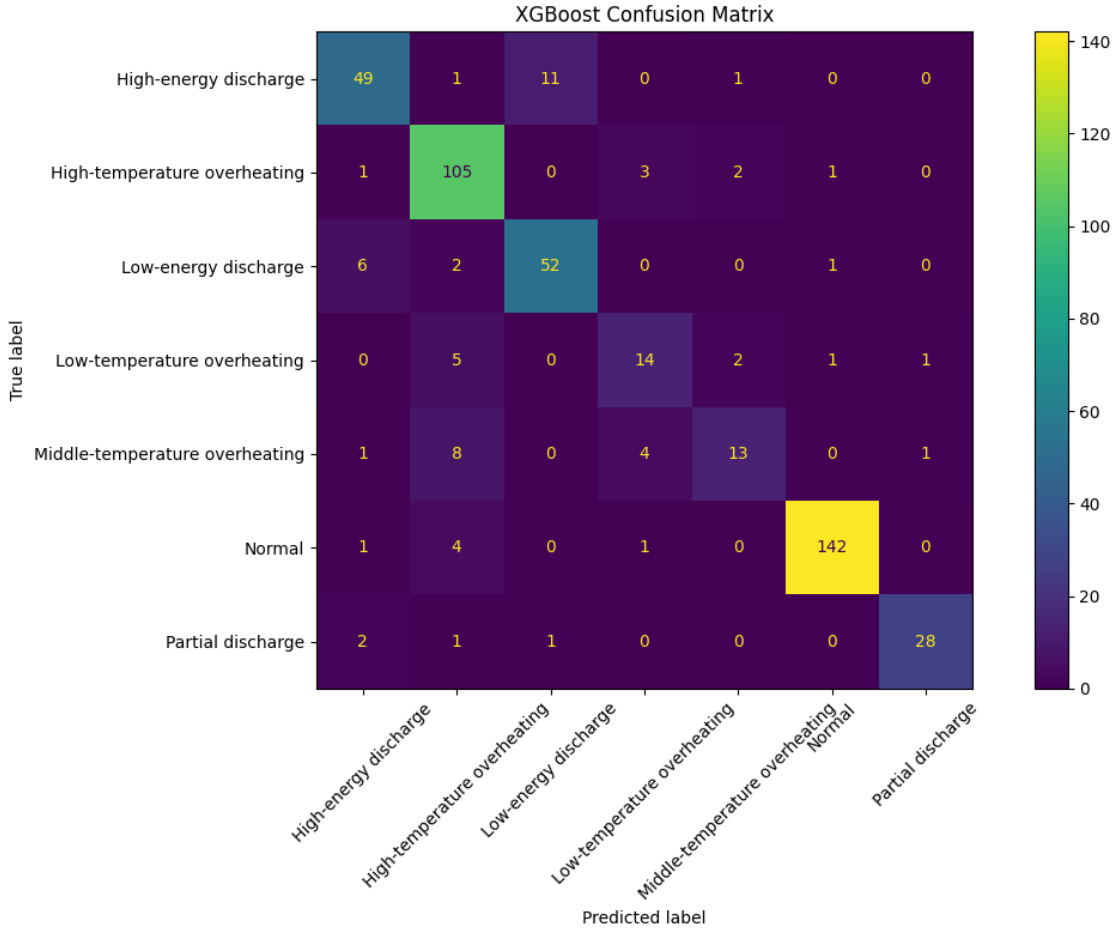


Figure 2: Confusion matrix for XGBoost.

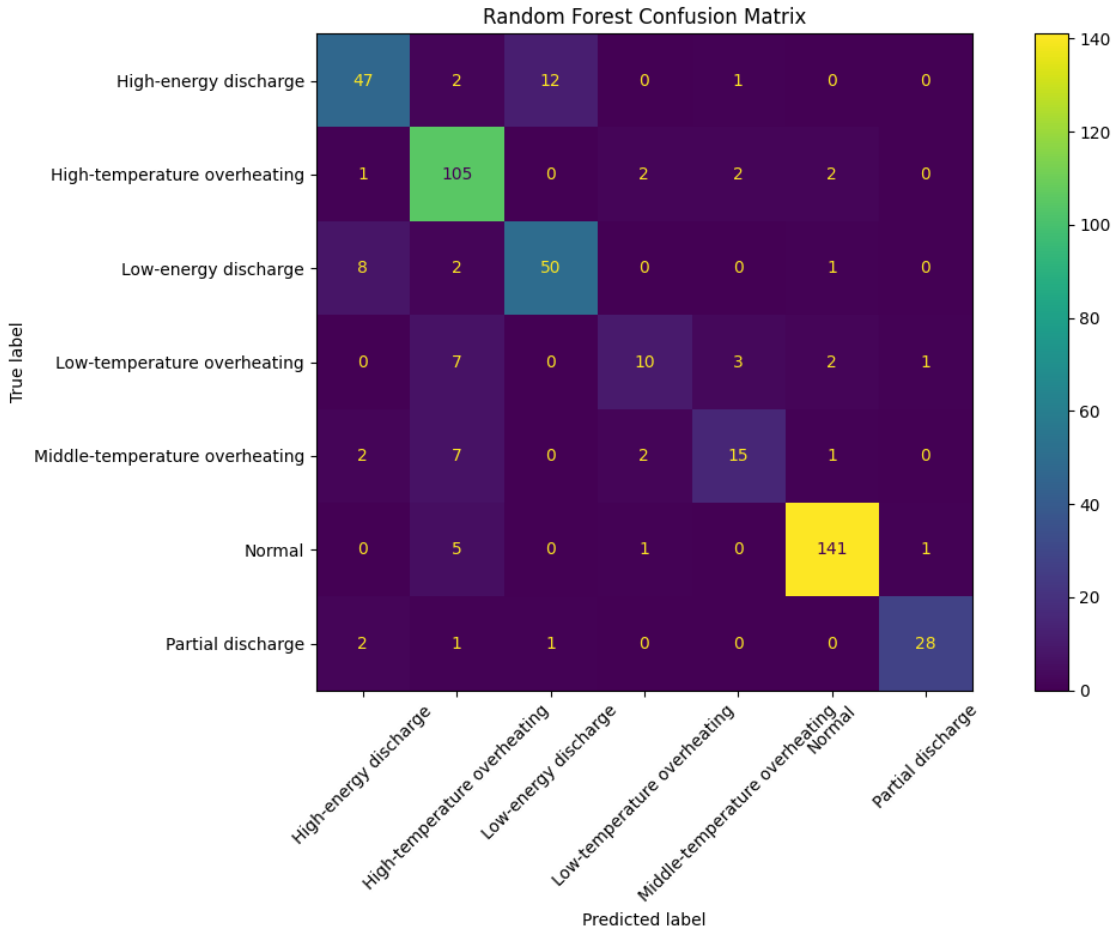


Figure 3: Confusion matrix for Random Forest.

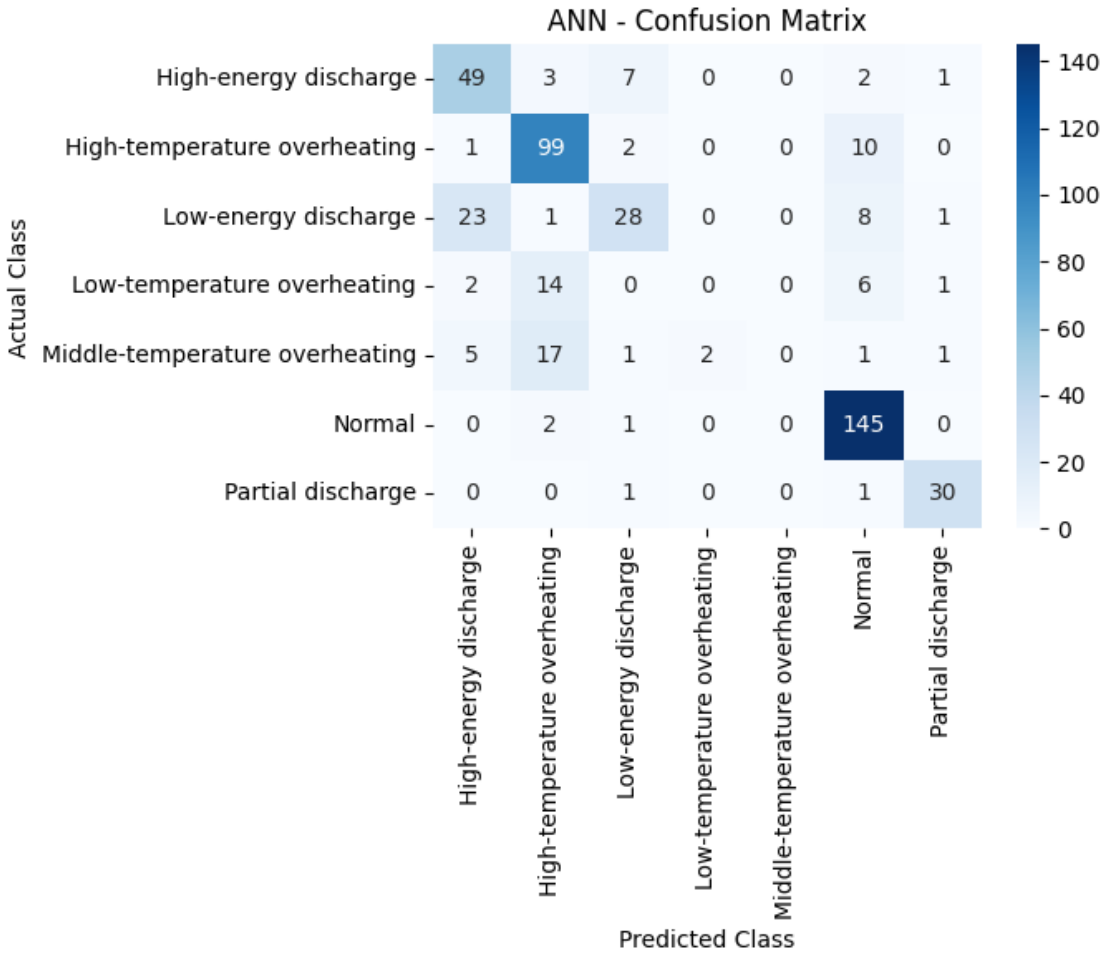


Figure 4: Confusion matrix for K-Nearest Neighbors.

5.4 Experimental Setup

The reported experiment uses an 80:20 training–testing partition. The evaluation compares the six classifiers using the same five-gas feature representation and seven target classes. The following results

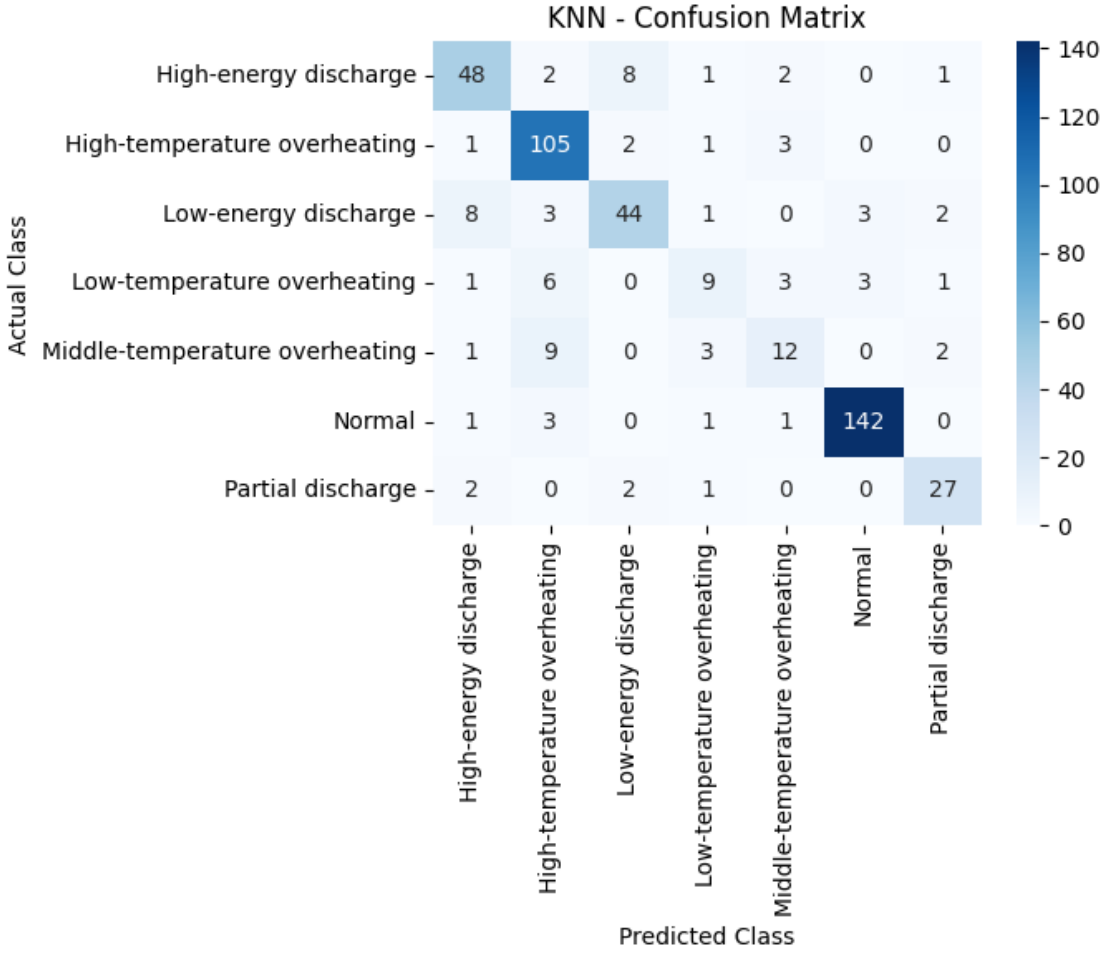


Figure 5: Confusion matrix for the Artificial Neural Network.

are transcribed from the supplied paper images; they should be checked against the original experiment logs before publication.

5.5 XGBoost

XGBoost achieved a reported accuracy of 86.67%, correctly classifying 403 of 465 samples according to the supplied

results. Its performance suggests that boosted decision trees can capture useful nonlinear relationships among the five dissolved-gas measurements. However, the result should be interpreted in the context of the dataset size, class distribution, and validation design.

5.6 Random Forest

Random Forest achieved a reported accuracy of 85.16%, corresponding to 396 correctly classified samples out of 465. The ensemble structure can model nonlinear interactions among gases and provides a

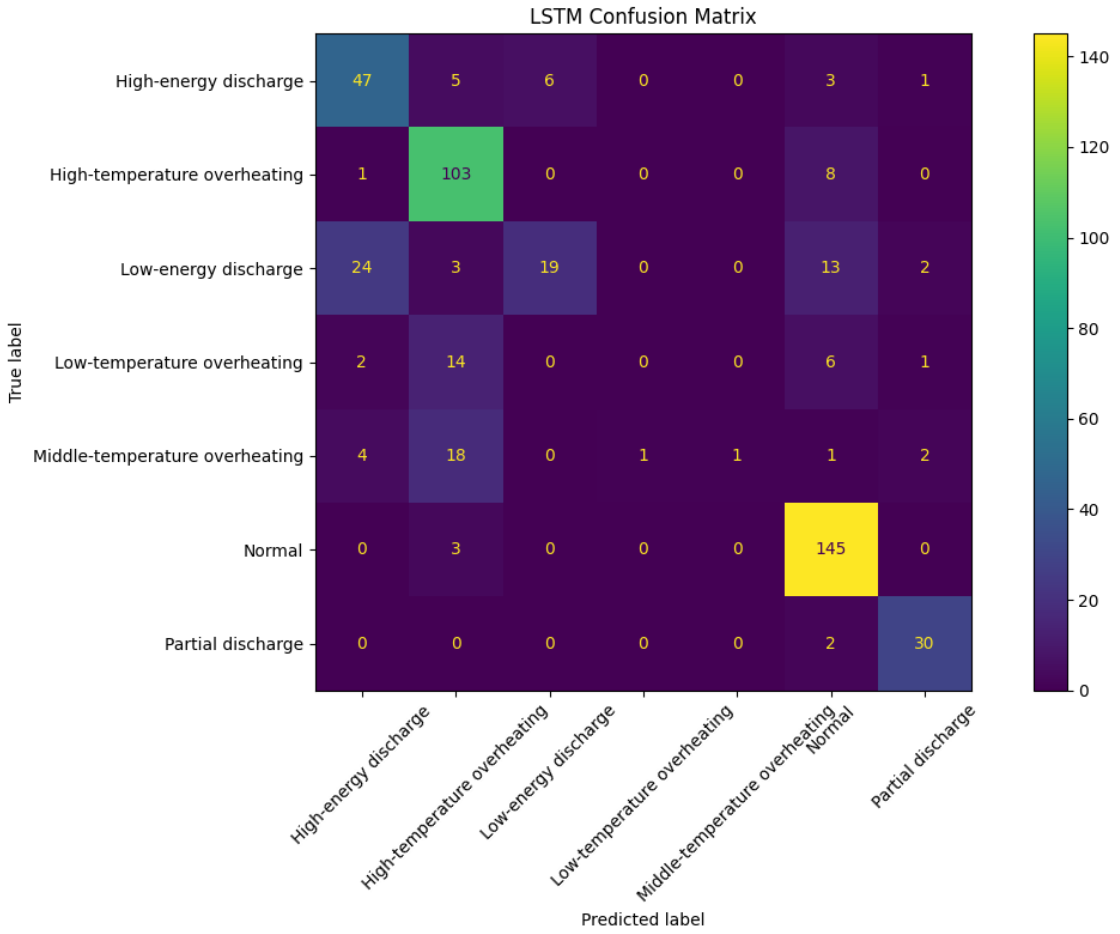


Figure 6: Confusion matrix for Long Short-Term Memory.

Table 2: Reported Performance of Machine-Learning Models

Model	Accuracy (%)	Precision (%)	Recall (%)	Specificity (%)	F1-score (%)	MCC
XGBoost	86.6667	86.5527	86.6667	97.7130	86.3757	0.832233
Random Forest	85.1613	84.8573	85.1613	97.4145	84.7137	0.812904
KNN	83.2258	82.4926	83.2258	97.1281	82.6484	0.788379
ANN	75.4839	67.6319	75.4839	95.5209	70.7157	0.689337
LSTM	71.8279	63.7857	71.8279	94.8707	60.6436	0.644656
SVM	57.6344	52.7251	57.6344	91.7062	51.8339	0.453024

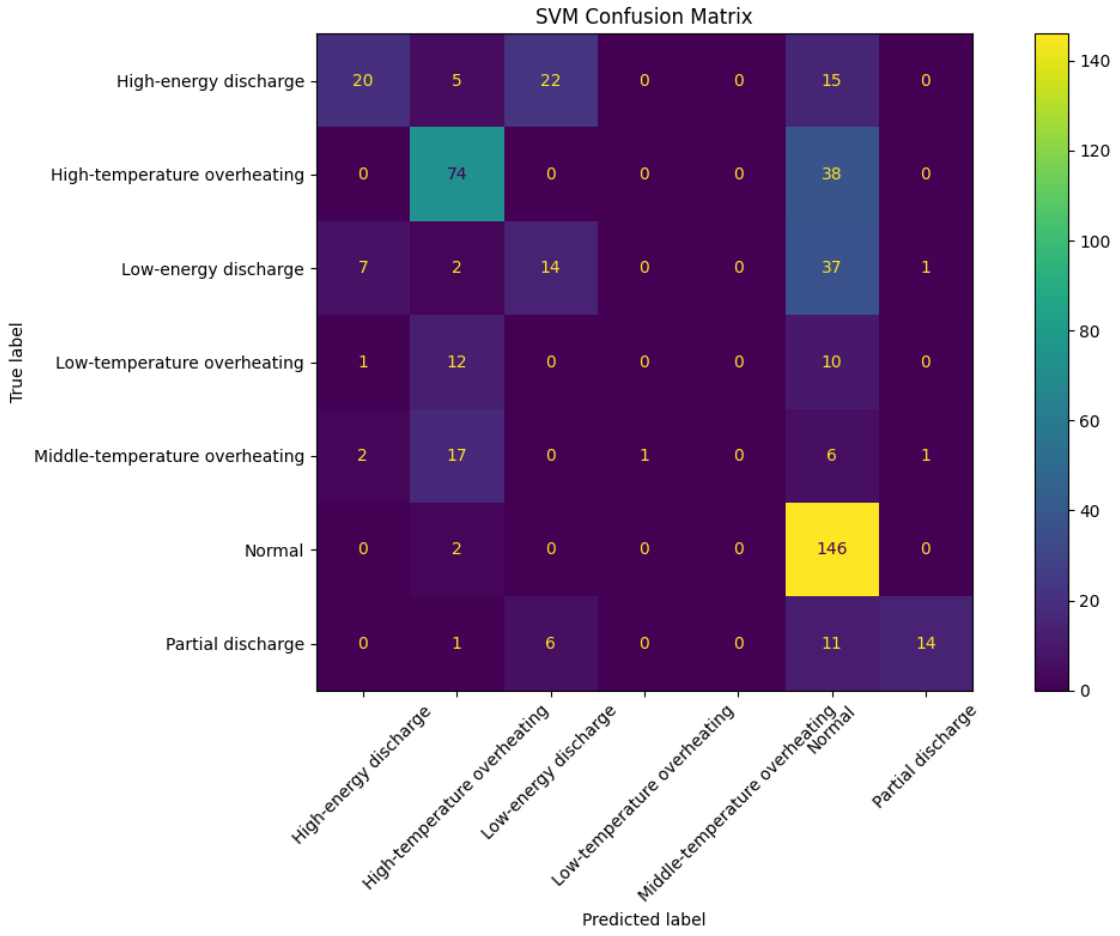


Figure 7: Confusion matrix for Support Vector Machine.

strong baseline for structured DGA data. The confusion matrix should be inspected to determine which transformer-condition classes are most frequently confused.

5.7 K-Nearest Neighbors

KNN achieved a reported accuracy of 83.23%. Its performance indicates that local similarity in the five-dimensional gas-feature space can be useful for distinguishing some transformer conditions. KNN remains sensitive to scaling, the selected neighbourhood size, and overlap between class distributions. H. ANN, LSTM, and SVM The supplied results report accuracies of 75.48% for ANN, 71.83% for LSTM, and 57.63% for SVM. These outcomes show that model complexity alone does not guarantee improved classification performance. ANN performance may be affected by the size of the dataset and network configuration. LSTM is most meaningful when temporal sequences are available; otherwise, its sequence-learning advantage may not be realised. SVM performance can depend strongly on kernel selection, scaling, and hyperparameter optimisation.

6 Comparative Interpretation

Among the reported models, XGBoost achieved the highest overall accuracy, followed by Random Forest and KNN. The choice of a final model should not rely on accuracy alone. Recall for each fault class, false-negative behaviour, confusion matrices, calibration, and performance on independent data should also be examined. In a maintenance setting, failure to identify a developing fault may have greater practical consequences than an isolated overall accuracy difference.

7 Predictive-Maintenance Framework

The proposed framework connects DGA acquisition, machine-learning classification, condition assessment, and maintenance decision support. It is intended to assist early identification and prioritisation of potentially abnormal transformer conditions rather than replace established protection systems or qualified engineering judgement.

7.1 DGA Data Acquisition

Dissolved-gas measurements are acquired from transformer oil. The five variables used in this study are H2, CH4, C2H6, C2H4, and C2H2. The measurements are checked for missing values, inconsistent units, invalid observations, and data-quality issues before being supplied to the model.

7.2 Data Quality and Feature Processing

The feature definitions and preprocessing procedures used during model development are applied consistently to new observations. If scaling or other transformations are required, the parameters established during model development are reused rather than recalculated using the test observation or future evaluation data.

7.3 ML-Based Fault Classification

The processed DGA feature vector is provided to the trained classifier. The classifier estimates the transformer condition among the seven classes considered in this study. The predicted class is treated as

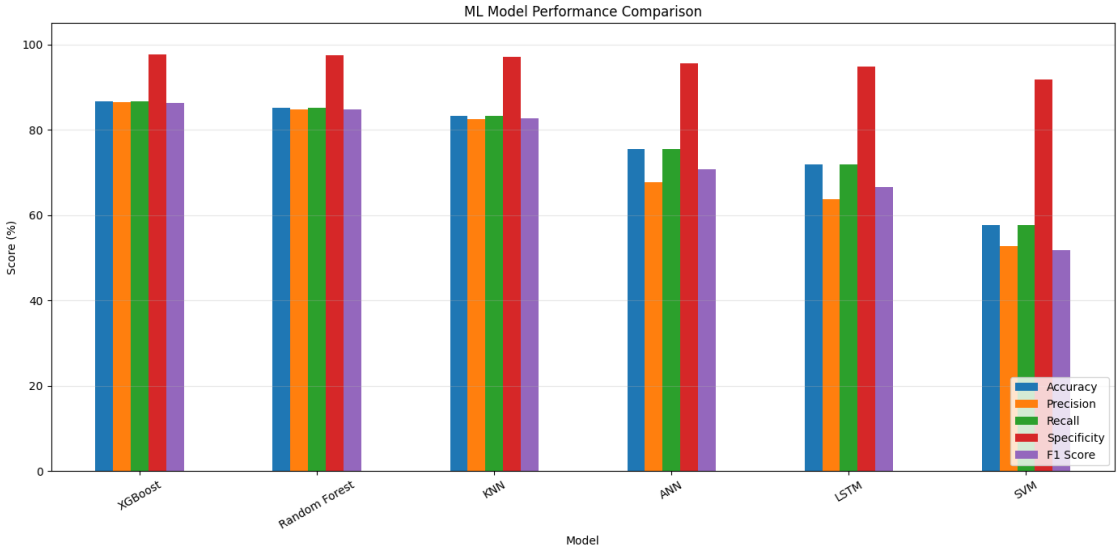


Figure 8: Comparison of reported evaluation metrics across the six models.

a diagnostic indicator and should be interpreted together with the measured gas values, transformer history, and other available evidence.

7.4 Transformer Condition Assessment

The predicted condition supports further assessment of transformer status. A normal-condition prediction may indicate that no abnormal pattern was identified by the model. A discharge or overheating prediction warrants appropriate diagnostic consideration. The model output is not, by itself, a definitive diagnosis or an automatic maintenance command.

7.5 Maintenance Decision Support

Maintenance priorities depend on the severity and operating context of the transformer condition. If an abnormal condition is indicated, the prediction can support further investigation, additional diagnostic testing, inspection, or maintenance planning. Historical DGA measurements, transformer loading, operating conditions, maintenance records, and other monitoring information should be considered before an intervention is selected.

7.6 Continuous or Follow-Up Monitoring

Where repeated DGA measurements are available, subsequent observations can be assessed using the same trained framework. Comparing predictions over time may help identify changes in transformer condition. However, temporal learning should only be claimed when the data contain genuine chronological information and the model is evaluated using an appropriate sequence-aware validation strategy.

8 Limitations And Future Work

8.1 Dataset Size and Representativeness

The performance of supervised machine-learning models depends on the size, variety, and representativeness of the training set. The dataset used in this study may not fully capture the diversity of real-world transformers, operating conditions, insulation materials, and DGA measurements. Accuracy values therefore should not be directly extrapolated to all field installations.

8.2 Limited Feature Representation

The model uses five dissolved gases. Transformer diagnosis may also benefit from other information, including carbon monoxide and carbon dioxide, gas-generation rates, gas ratios, DGA trends, oil temperature, transformer load, and other condition indicators. Additional features should be selected and validated according to their availability and engineering relevance.

8.3 Ambiguous Fault Signatures

Different fault mechanisms may produce similar gasconcentration patterns, particularly for related discharge and overheating categories. The five-dimensional feature space may not be sufficient to separate every class, and some misclassifications may arise from inherent overlap between observed gas signatures.

8.4 Data and Label Reliability

Supervised learning depends on the accuracy of the target labels. For real-world transformers, labels may be obtained from diagnostic tests, maintenance reports, or other sources. Variations in measurement quality, equipment calibration, missing labels, and differences in labelling practice can affect model reliability.

8.5 Limited Temporal Dynamics

DGA measurements are often collected repeatedly during transformer operation. The temporal sequence may contain information about fault development, but the current feature representation focuses on individual observations. Future work can investigate sequence-based models when chronological measurements are available.

8.6 Independent Validation and Interpretability

The models should be validated using independent field datasets collected from different transformers, manufacturers, operating conditions, and measurement systems. Modelinterpretation techniques can help establish whether important predictions are consistent with known electrical and thermal fault mechanisms. Confidence estimation and uncertaintyaware decision support may help identify observations that require additional investigation.

8.7 Future Work

Future work should focus on: 1) collecting larger and more representative DGA datasets from multiple sources; 2) validating the models on independent field data and alternative measurement systems; 3)

evaluating temporal models using genuinely sequential DGA measurements; 4) exploring additional gas features, gas ratios, transformer loading, and operating conditions; 5) improving interpretability and uncertainty estimation; and 6) assessing practical integration with condition-monitoring and maintenance workflows.

9 Conclusion

This study examined machine learning for DGA-based fault classification and predictive maintenance of oil-immersed power transformers. Five dissolved gases were used as input features to six classifiers: Random Forest, XGBoost, KNN, ANN, SVM, and LSTM. The reported comparison showed that XGBoost achieved the highest accuracy at 86.67%, followed by Random Forest at 85.16% and KNN at 83.23%. These findings suggest that machine learning can provide useful support for interpreting DGA patterns and identifying transformer operating and fault conditions. Nevertheless, the results depend on dataset quality, class representation, preprocessing, and validation design. Independent field validation, larger datasets, and careful engineering interpretation are necessary before the framework can be relied upon in practical transformer maintenance.

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